Confidence-Based Abstention for Biomedical Citation Verification: A Retrospective Evaluation of Archived Predictions

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Abstract

A reference can be genuine and still be used to support a claim that the cited paper never made. Automated citation checkers face the same difficulty as human readers: a plausible interpretation is not necessarily a faithful one. This study examines a modest way of handling uncertainty in such systems. Rather than changing the citation verifier, we use its archived class probabilities to decide which predictions to retain and which to leave for review. The analysis covers 606 biomedical claim–document pairs. The original classifier made 268 errors, an error rate of 44.22%. Retaining the 485 highest-confidence predictions (80.03% coverage) left 195 errors, or 40.21%. This is 4.02 percentage points below the error expected from accepting the same number of predictions at random; a pair-level bootstrap gave a descriptive 95% interval of 1.91–5.91 points. The improvement has an important qualification: F1 for `NOT_ACCURATE` fell from 0.3123 to 0.2788 in the retained subset. Excluding 50 cases with provisional labels still gave a 3.39-point reduction in overall error. Confidence ranking therefore helps identify an error-rich set of cases to withhold, but it does not reliably isolate the misleading citations that matter most. The analysis is retrospective, and unresolved questions about the original model's data splits and label mapping prevent claims about deployment on unseen articles.

Keywords

citation integrity, scientific claim verification, selective prediction, abstention, risk–coverage, uncertainty

1 Introduction

A citation is often read as a promise: the referenced paper should provide evidence for the statement beside it. Yet a correct reference is not the same as a correct account of what that reference says. A statement may drop an important limitation, generalize a result beyond the population studied, or describe an association as if it established causation. None of these problems is fixed by checking that the title, author, and DOI are real. The issue is whether the citing text represents the source faithfully.

This distinction matters particularly in scientific writing, where a claim may be repeated long after its original context has been forgotten. Greenberg's account of a biomedical citation network illustrates how repeated citation can give unsupported claims an appearance of authority [3]. Such errors are not necessarily deliberate. They can also arise from compressed summaries, ambiguous language, or failure to read the qualification attached to a finding. Whatever the cause, checking citation fidelity requires examining a claim against the evidence in a known source.

There is now enough annotated material to test automated approaches to this task. Sarol *et al.* studied 3,063 biomedical citation instances referring to 100 source articles and evaluated retrieval-based citation verification [5]. Their strongest reported top-20 evidence configuration obtained 0.59 micro-F1 and 0.52 macro-F1. Those scores establish that the task is computationally approachable, but also show that the models still make substantial mistakes. Accuracy alone is not the whole story. An erroneous prediction that a misleading citation is accurate could be more consequential than an unnecessary warning on a faithful citation.

The practical question behind this paper came from that observation. If a verifier is uncertain, must it issue a definitive label? One alternative is to reserve a portion of predictions for manual review. In selective classification, the fraction of predictions retained is called *coverage*; the error rate among them is *selective risk* [2]. Reducing coverage is useful only if the cases withheld contain a disproportionate share of the mistakes. Randomly discarding one-fifth of the cases, for example, does not lower the expected error rate among those that remain.

We examine that idea using predictions saved from an existing biomedical citation classifier. The study does not train a model, alter its architecture, or introduce a new abstention algorithm. It asks three narrower questions. First, does a maximum-probability ranking yield lower error than random retention when both accept 80% of cases? Second, how does the trade-off change as coverage varies? Third, and most importantly for citation auditing, what happens to errors involving the `NOT_ACCURATE` class?

A saved-prediction analysis is also a pragmatic choice. Earlier attempts to run the original Longformer-based verifier required an older software environment and substantial Colab memory. The archived probabilities let us evaluate the decision policy on CPU without confusing implementation changes with changes in the model's predictions. We can reproduce the numerical comparison, although we cannot claim a fresh end-to-end replication of the original neural training procedure.

1.1 Why the choice of metric matters

One reason for examining abstention is that accuracy hides how decisions are distributed. Consider two systems that each make 268 errors on 606 citations. If one is uncertain about most of its mistakes and the other is extremely confident about them, they have identical accuracy but offer different possibilities for triage. A confidence ranking does not repair either system. What it can do is separate some decisions that might safely be handled automatically from others that deserve closer attention. We examine that separation directly instead of treating a single F1 score as a complete account of reliability.

There is a second distinction that becomes more important in citation checking than in many standard classification exercises. The three labels do not carry the same practical meaning. A false

NOT_ACCURATE label may create an unnecessary editorial query. An inaccurate citation labeled ACCURATE, on the other hand, might pass without scrutiny. Our experiment has no measured costs for either event, so we do not turn this intuition into an invented utility score. We retain the ordinary overall-error metric to keep the policy comparison interpretable, but also report what happens specifically to the inaccurate citations. If the two tell different stories, the disagreement belongs in the findings rather than being averaged away.

This paper accordingly addresses a limited decision rather than every aspect of citation trustworthiness: given a fixed set of predictions and a fixed review budget, does maximum-probability ranking identify a subset that contains fewer mistakes than a random subset of equal size? The question is small enough to answer with available evidence. It also makes the boundary of the contribution visible. We do not test whether an editor would notice a missing qualification in the underlying paper, and we do not infer that an accepted citation is scientifically correct in any wider sense.

2 Related Work

2.1 From citation context to supporting evidence

Citation analysis has addressed several questions that should not be conflated. Citation-intent classification, for instance, asks why an author cites a paper: to describe background, use a method, or contrast findings. Cohan *et al.* showed the value of document structure for classifying such roles [1]. That is useful information, but a method citation can still misdescribe the method, just as a background citation can exaggerate a result. Our concern is the factual relationship between the citing sentence and the referenced source.

Scientific fact verification offers a closer starting point. SciFact links scientific claims to evidence in research abstracts and includes sentence-level rationales [6]. MultiVerS uses full-document context and weak supervision for scientific claim verification [7]. These systems connect a label to source evidence rather than treating a claim as ordinary text classification. Sarol *et al.* adapted retrieval and MultiVerS-style verification to biomedical quotation errors [5]. Their labels are ACCURATE, NOT_ACCURATE, and IRRELEVANT. With five retrieved sentences, the published system reports 0.58 micro-F1 and 0.50 macro-F1; with 20 sentences it reports 0.59 and 0.52. We use those figures for context, not as a directly controlled comparison with our archive, whose checkpoint and processing provenance have not been fully recovered.

2.2 Choosing not to predict

Selective classification makes abstention an explicit part of evaluation rather than an afterthought [2]. A model can accept fewer predictions and reduce its error on the accepted portion, but that reduction has to be examined alongside the decisions it declines to make. Probability calibration is a different question. Guo *et al.* studied how confidence scores can be adjusted to better reflect observed correctness [4]. We make no calibration claim here. Our only use of the three saved probabilities is to order cases by their largest value. In particular, a confidence score of 0.9 is not assumed to mean that the prediction is correct 90% of the time.

The difference is consequential for this dataset. Because we cannot establish that the archived development partition was independent of training, fitting a new temperature or selecting a confidence threshold on it would be difficult to defend. We therefore ask how well the existing scores *rank* cases in a fixed archive, not whether they provide a deployment-ready probability of correctness.

3 Methodology

3.1 What was available in the archive

We work with one claim and its cited document at a time. Each record contains a claim identifier, one cited-document identifier, the predicted class, and three probabilities ordered as NOT_ACCURATE, NEI, and ACCURATE. For claims with explicit evidence labels, the reference label supplies the gold class. A record with an empty evidence field is assigned NEI *provisionally*; an empty field does not by itself establish that an annotator explicitly marked the citation IRRELEVANT. There are 606 test pairs, including 50 such cases, and a separate archive of 316 development pairs.

The records were checked before analysis. We matched each prediction to the corresponding claim and cited document, looked for duplicate keys, verified that the three probabilities were finite and summed to approximately one, and checked that the saved predicted label matched the highest probability. File hashes were recorded so that later runs could be compared with exactly the same input files. The development data were used for these structural checks, not for choosing a new threshold.

3.2 The acceptance rule

For case i, let $\mathbf{p}_i$ be the vector of class probabilities, $\hat{y}_i$ the stored prediction, and y_i the analysis label. The score used for ranking is

$$s_i = \max_c p_{ic}, \qquad z_i = \mathbb{I}(\hat{y}_i = y_i), \tag{1}$$

where z_i records whether the prediction is correct. We sort test cases by s_i, from highest to lowest, using identifiers and original row order to break ties. At a target coverage q, the number accepted is

$$k(q) = \lfloor qN + 0.5 \rfloor . \tag{2}$$

For the accepted set A_q, coverage and error among accepted predictions are

$$C(q) = \frac{|A_q|}{N}, \qquad R(q) = \frac{\sum_{i \in A_q} (1 - z_i)}{|A_q|}. \tag{3}$$

An abstention is neither a correct classification nor an error in the numerator; it is a decision the policy does not issue. At the primary target $q = 0.80$, nearest-integer rounding retains 485 of 606 cases, giving 80.03% realized coverage.

An important feature of this rule is what it *does not* do. It does not change a predicted class, inspect the gold label while ranking, or use retrieved sentences to override the classifier. The only intervention is deciding which of the archived predictions to accept. In our fixed test batch, selecting the top k also guarantees the number retained. That would not be equivalent to setting a confidence threshold on an unseen stream of manuscripts.

3.3 Controls and the quantity we compare

We examine three policies: retaining every prediction, retaining k uniformly sampled cases, and retaining the k cases with the largest scores. Uniform sampling is a useful reference because every prediction has the same chance of being kept. Consequently, its expected selective error is the error in the full pool:

$$\mathbb{E}[R_{\text{random}}(q)] = R(1). \tag{4}$$

The primary effect is expressed in percentage points:

$$\Delta_{80} = 100\,[R(1) - R(0.8)]. \tag{5}$$

We also ran 1,000 random-retention draws to check the expectation empirically. They are repeated selections from the same records, not independent model runs.

For clarity, the 80% policy produces four groups: 290 accepted/correct, 195 accepted/incorrect, 48 rejected/correct, and 73 rejected/incorrect. All four matter. Reporting only the higher accuracy of accepted predictions would conceal the cost of withholding 48 decisions that the classifier had actually made correctly.

3.4 Uncertainty, sensitivity, and reproducibility

To describe sampling variation within this archive, we used 2,000 bootstrap samples of claim–document pairs. For each sample, both the full-pool error and the top-k error were recalculated, including the ranking step. The middle 95% of these differences, bounded by the 2.5th and 97.5th percentiles, forms a descriptive interval. This is not an article-clustered interval and is not a test of performance on every possible future citation.

We repeated the analysis at 70%, 90%, and 100% coverage and examined class-wise F1 and recall. We also reran the 80% comparison after removing 50 examples with provisional labels. Margin and entropy rankings were explored separately; because those alternatives were examined using the same test records, they are not treated as a newly validated choice of method. The random-retention seed was 42, and the bootstrap seed was 2026. The evaluation was performed on CPU using Python, NumPy, pandas, Matplotlib, and scikit-learn.

3.5 Auditing selection rather than reclassification

It helps to view the decision policy as a short audit trail. First we verify that a pair has exactly one archived class prediction and three scores. Next we order the pairs by the largest score. We then mark the first k records as accepted and every later record as withheld. Finally, after selection, we compare the original labels with the analysis labels. The policy never searches a new document, rewrites a claim, or issues a replacement class.

This distinction keeps the experiment controlled. If we had also changed retrieval length, tokenization, or checkpoint weights, an improvement might have come from one of those changes rather than from abstention. Using the same archive across full acceptance, random retention, and confidence retention lets us isolate the value of the ordering. The comparison is still conditional on that particular verifier: it says nothing about how a different architecture would behave until such a system is tested under the same protocol.

A practical detail is the difference between *rejection* and *correction*. When a prediction is withheld, the automatic process no longer makes that decision, but the underlying citation has not been verified. A manual reviewer might resolve it, ignore it, or reach a different judgment. We observe none of these later events. For this reason the reported reduction is explicitly an error rate among *accepted automatic decisions*. Counting a rejected case as correct would make almost any abstaining system look good, even one that simply refused to answer difficult questions.

4 Experimental Setup

4.1 Data, checkpoint history, and comparison limits

The recovered inputs were four JSONL files: gold claim records and saved predictions for development and test. The test set contains 386 `ACCURATE` cases, 170 `NOT_ACCURATE` cases, and 50 cases assigned provisional `NEI`. All 922 development and test records matched their predictions. We did not rerun the document encoder or refit any neural weights for this study.

This distinction is especially important in interpreting the baseline. The checkpoint metadata point to a Longformer-based three-class verifier with five training epochs. They also report 2,454 training instances, matching the earlier counts for training and development combined. That agreement raises the possibility that the nominal development examples were used during training. We also cannot rule out use of the test partition during checkpoint validation. Neither issue is resolved by the present data. The evaluation reported here is therefore a *retrospective analysis of an archived predictor*, not a clean evaluation of a newly trained model on an untouched test set.

The local archive yields 0.5578 micro-F1 and 0.4438 macro-F1 at full coverage. Sarol *et al.* report 0.58 and 0.50 for a published top-five configuration [5]. We include the difference as a reproducibility warning: checkpoint identity, retrieval settings, preprocessing, and exact label mapping have not been matched closely enough for a formal head-to-head comparison.

4.2 Evaluation decisions

We focus on the 80% accepted set because it was the documented primary setting in the later experimental protocol. The practical reference was an improvement of at least three percentage points, accompanied by a bootstrap interval with a positive lower bound. An earlier exploratory notebook had already examined related confidence analyses before this protocol was written, however. We therefore treat the criterion and its interval as descriptive, not as evidence of prospective preregistration. Additional checks address the coverage trade-off, class-specific errors, and the provisional-label subset. Plots show 70–100% coverage for readability; the complete ranking curve remains in the saved outputs.

4.3 Why no new model training was needed

The earlier notebook was valuable mainly because it left behind the quantities needed for a controlled analysis: the original class probabilities and the model's decisions. In addition to saving computation, this preserves a meaningful baseline. Training a second classifier or retuning a checkpoint after seeing test-set errors would change the question and introduce new sources of variation. We instead

Table 1: Selective error at different acceptance levels on 606 archived test pairs. Random retention has 44.22% expected error at each level.

Nominal coverage	Accepted k	Errors	Risk (%)	Gain (pp)
70%	424	168	39.62	+4.60
80%	**485**	**195**	**40.21**	**+4.02**
90%	545	230	42.20	+2.02
100%	606	268	44.22	0.00

use the saved predictions as fixed inputs and keep all subsequent changes inside the decision policy and evaluation scripts.

The approach has a clear cost as well. We cannot inspect the model's intermediate attention patterns, change the evidence shown to it, or determine whether an error began in retrieval or in the final decision. Nor can we infer the training-time memory cost of the encoder from a quick CPU evaluation of stored probabilities. These restrictions are recorded rather than filled with assumptions. The experiment should be understood as a reproducible analysis of *outputs* from a model, with the original training configuration providing context but not a fully reconstructed training pipeline.

5 Results

5.1 What changed at 80% coverage?

Without abstention, 338 of 606 predictions are correct and 268 are wrong. This is 55.78% accuracy, or 44.22% error. The confusion matrix in Table 3 already points to a serious limitation: 102 of 170 citations labeled NOT_ACCURATE were predicted ACCURATE. A system intended to help readers notice distorted citations can miss them even when its overall accuracy appears reasonable.

Selecting the 485 highest-confidence predictions gives 195 errors, an accepted-set error rate of 40.21%. Randomly accepting 485 would leave the expected error at 44.22%; the average over 1,000 random draws was approximately 44.20%. The observed gain is therefore 4.02 percentage points (Eq. 5). The descriptive pair-level bootstrap interval runs from 1.91 to 5.91 points. These figures describe the same fixed archive. They do not show that the policy would produce that improvement for an unrelated set of articles.

The withheld group helps explain the improvement. Of 121 rejected predictions, 73 are errors and 48 are correct. The rejected group's error rate is 60.33%, whereas the complete pool's is 44.22%. In other words, rejecting about one-fifth of the predictions removes 27.24% of all errors, but it also removes 48 correct labels. That second number is part of the result, not an inconvenience to be hidden.

5.2 Coverage is a trade-off, not a free gain

The results at other coverage levels are shown in Table 1 and Fig. 1. With 70% of predictions accepted, error is 39.62%. Accepting 90% raises it to 42.20%, and accepting everyone returns it to 44.22%. The curve has local ups and downs, as one would expect when individual predictions enter a ranked list. Its discrete full-prefix area is 0.35373, but very small prefixes have unstable error rates; our main interpretation rests on the more substantial 80% group.

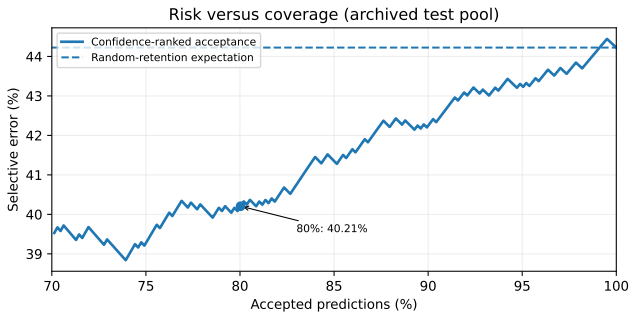


Figure 1: Error among accepted predictions as coverage increases. The dashed line is the expected error from uniform random retention at the same accepted count.

Table 2: Class-wise F1 on the full pool and on the 80%-accepted subset. The subsets differ; these numbers do not reflect updated model weights.

Gold class	Full set F1	Accepted F1	Accepted support
ACCURATE	0.6933	0.7396	321
NOT_ACCURATE	0.3123	0.2788	125
NEI*	0.3259	0.3717	39
Macro-F1	0.4438	0.4634	485

*Gold labels are provisional for empty evidence records.

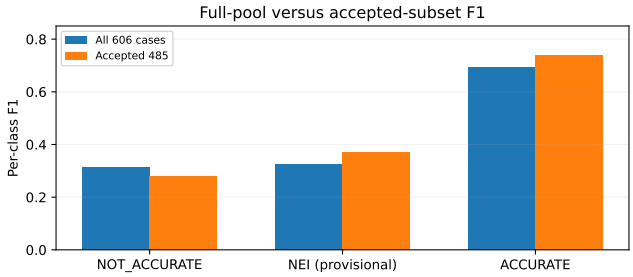


Figure 2: Class-wise F1 before and after selection. Overall selective error improves, but the NOT_ACCURATE F1 score falls.

5.3 The difficult class remains difficult

The overall macro-F1 rises from 0.4438 on all 606 cases to 0.4634 on the 485 accepted cases. It would be misleading, however, to describe this as improved model performance without looking at the class-wise numbers. The model itself has not changed, and the two F1 scores are computed on different populations. As Table 2 shows, F1 for ACCURATE and provisional NEI increases, while F1 for NOT_ACCURATE *decreases*, from 0.3123 to 0.2788.

The counts show what this means in practice. Of 125 retained NOT_ACCURATE cases, only 29 are correctly identified; 77 are labeled ACCURATE and 19 are labeled NEI. Recall for inaccurate citations is 47/170 (27.65%) in the full set and 29/125 (23.20%) after selection. At least 75 errors in the complete archive have maximum probabilities

Table 3: Full-pool test confusion matrix. Rows are analysis labels; columns are predictions.

Gold class	Predicted N_ACC	Predicted NEI	Predicted ACC
NOT_ACCURATE	47	21	102
NEI (provisional)	9	22	19
ACCURATE	75	42	269

of 0.90 or more. These are not low-confidence mistakes that an ordinary reject option will reliably catch.

Selection is not equally distributed across gold classes. It retains 321 of 386 ACCURATE cases (83.2%), 125 of 170 NOT_ACCURATE cases (73.5%), and 39 of 50 provisional NEI cases (78.0%). These are retrospective class-conditional rates, not the output of a classifier that had access to the true class at selection time. They help explain why the mix of accepted cases can change F1 even though no prediction label is altered.

5.4 How the decision count changes by class

The accepted-set composition deserves attention because an overall error reduction can coexist with weaker performance for an important class. In the complete test set, ACCURATE is the largest category (386 of 606 cases). After confidence-based selection, the corresponding retained support is 321. By comparison, NOT_ACCURATE falls from 170 to 125, and provisional NEI from 50 to 39. These are not the same retention percentages. As a result, neither accepted-set accuracy nor accepted-set macro-F1 is measuring exactly the same distribution as the full baseline.

There are two separate questions here. Was a difficult case withheld? And if it was accepted, was its label correct? The confidence rule addresses the first without directly solving the second. It can reject many mistakes overall while leaving mistakes concentrated in a particular label relationship. For citation auditing, the conspicuous relationship is NOT_ACCURATE being predicted ACCURATE. The full confusion matrix contains 102 such mistakes, and the accepted subset still contains 77. The policy has reduced their count, but 77 remain as automatic assertions of accuracy. This is a reason to report the absolute counts alongside class-wise F1, rather than relying entirely on the percentage-point headline.

5.5 Does the provisional label mapping drive the result?

There is a known weakness in the local labels: 50 test records have empty gold-evidence fields and were treated as NEI. We therefore repeated the analysis on the 556 records with explicit gold labels. Their full-pool error is 43.17%. Retaining the 445 most confident cases gives 39.78% error, a reduction of 3.39 percentage points. The direction of the main finding persists, although the reduction is smaller. This check does not establish that every excluded case should have been labeled IRRELEVANT; that question needs an audit of the original annotations.

6 Discussion

6.1 What the confidence scores tell us—and what they do not

The simplest reading of the experiment is that the scores carry some information about error. Correct predictions have mean maximum probability 0.8118, compared with 0.7434 for incorrect predictions. Consequently, the bottom-ranked fifth contains a higher concentration of errors than the full test pool. This is useful evidence for ranking, even though it is far from a guarantee that the remaining predictions are correct.

It is worth separating this finding from a stronger claim about a real editorial workflow. The data show which archived predictions would be withheld. They do not measure how much time a reviewer would spend checking those citations, whether a reviewer would resolve the disagreement correctly, or how many important misrepresentations would be missed despite high model confidence. Sending rejected cases to a person may be sensible, but the present experiment does not establish the value or cost of that workflow.

There is also a difference between a fixed review quota and a deployed system. Here we already know the batch size: 606 cases. We can sort them and accept exactly 485. A journal processing new submissions one by one does not have this future batch available. To choose a numerical threshold prospectively, an independent and representative calibration set would be needed. For the same reason, the numbers cannot be used to claim that the model's probability values are calibrated.

6.2 Why the negative class result matters

The disappointing NOT_ACCURATE result is not incidental. It is close to the motivation for building citation-integrity tools in the first place. A system that often tells a reviewer that a misleading citation is accurate may provide false reassurance, even if the system's average accepted-set accuracy improves. In this archive, 77 inaccurate citations are still labeled accurate among the 485 retained predictions. The cost of that failure could be quite different from the cost of mistakenly flagging an accurate citation, but we have not estimated those costs.

Why does the model remain confident about such mistakes? Several explanations are plausible: it may be reacting to lexical overlap instead of the precise claim; the retrieval stage may have missed a qualifying passage; or an annotation may itself be ambiguous. Our saved probabilities cannot distinguish among these possibilities. That would require reading the original citing sentence and the corresponding scientific evidence for a targeted sample of high-confidence mistakes. We therefore do not infer a mechanism from the F1 values alone.

A class-sensitive acceptance rule is a possible follow-up, but it would need clear goals. For example, a study might require a minimum recall for inaccurate citations while limiting review workload. Thresholds should be learned on a genuinely independent development set, and success should be judged on articles not previously used to select the rule. Optimizing a new rule on the same 606 cases would produce a more flattering number without supplying better evidence.

6.3 How much does the uncertainty interval mean?

Our bootstrap interval describes variation when the 606 archived pairs are resampled and re-ranked. It is a useful check on whether the observed 4.02-point difference depends entirely on a handful of cases. Yet pairs drawn from the same reference article may share errors, so treating them as independent can understate uncertainty. The interval also says nothing about variation across model training seeds, new publishers, or other disciplines. Its positive lower bound is reassuring within the analysis performed, but is not proof of generalization.

Finally, random rejection is a deliberately simple control. It answers whether the confidence ranking is more informative than choosing the same number of cases without looking at confidence. It does not establish superiority over a different verifier or a trained selective-prediction model. Exploratory rankings by prediction margin and entropy gave 80%-coverage errors of about 41.03% and 39.59%, respectively. We report them as leads for future work, not as evidence that the apparent best alternative has been independently validated.

6.4 What a stronger next experiment would require

The next useful step is not simply to try another threshold on the same test records. An independently held-out collection, preferably split at the level of source articles, would answer the more basic question of whether the observed ordering survives new evidence. A proper development set could then be used to select a numerical confidence threshold without looking at the final evaluation cases. That design would distinguish a fixed quota, such as the one used here, from a policy that processes citations one at a time.

A second experiment should examine a manageable sample of the confident errors, starting with cases labeled `ACCURATE` when the gold label is `NOT_ACCURATE`. For each case, a researcher would need the citing claim, the source passage, and the annotation instructions. The review could then distinguish at least three possibilities: evidence retrieval missed the decisive sentence; the retrieved passage was available but misunderstood; or the annotation does not settle the issue unambiguously. Our current CSVs do not contain enough validated evidence for that diagnosis. It would be inappropriate to assign frequencies to those possibilities without doing the manual work.

Finally, a deployment-oriented study needs a meaningful measure of editorial cost. A journal may prefer to flag every likely inaccurate citation, even at the expense of more false alarms. Another workflow may have little capacity for review. The optimal policy depends on these constraints. Reporting reviewer time, false reassurance, and missed misleading citations at matched review budgets would make the outcome more relevant than optimizing average accuracy alone. These are directions for subsequent work, not claims established by the present experiment.

7 Limitations and Threats to Validity

Uncertain model provenance. The source files and archived model metadata do not establish that the nominal test set was untouched when the original classifier was developed. The training count also suggests possible inclusion of development records. We can reproduce the evaluation of stored probabilities, not the full model training or an independent test of a newly selected system.

Results examined during exploration. An earlier notebook had already explored calibration and abstention before the later protocol was written. The three-point target and bootstrap interval should therefore not be presented as if they were preregistered before anyone looked at the data.

Provisional labels. Empty gold-evidence records were mapped to `NEI`, whereas the published corpus uses an `IRRELEVANT` category. The explicit-label analysis reduces dependence on that convention, but it does not verify the missing original annotations.

Related examples and a single archive. Pair-level resampling ignores article-level clustering. We did not examine independent model seeds, changes in document retrieval, or performance in other research domains. Nor did we measure the original encoder's resource use in the present CPU analysis.

Different populations after abstention. The accepted group is selected using model scores. Its class mix differs from that of the full dataset, so F1 values before and after selection are diagnostic descriptions rather than a controlled change in model capability.

8 Reproducibility and Research Integrity

The reproducible object in this study is the archived prediction set and the code that evaluates it. The project retains SHA-256 checks for its four source files, the per-case acceptance decisions, the 1,000 random samples, 2,000 bootstrap differences, full risk–coverage outputs, class-wise summaries, and the scripts used to produce the figures. The seeds are 42 for the random control and 2026 for bootstrapping. Whether derived datasets and model assets can be publicly redistributed under the original licenses should be checked before release [5].

AI tools were used during debugging, code preparation, and drafting. The numerical findings in this paper come from recorded executions and files rather than generated examples. The earlier high-memory Colab environment and subsequent lighter CPU workflow are documented in the project log. The research paper is available in the project's GitHub repository:

https://github.com/mai2026011/Awesome-Detecting-misinterpreted-but-genuine-citations

The complete reproducibility package, including experiment scripts, configurations, and evaluation outputs, will be verified and documented separately.

9 Conclusion

The archived classifier's confidence scores were useful, but only within a limited sense. Keeping its 485 most confident decisions out of 606 reduced error from 44.22% to 40.21%, outperforming random retention at the same coverage by 4.02 percentage points. The price was withholding 121 decisions, including 48 correct ones. More seriously, detection of inaccurate citations did not improve: `NOT_ACCURATE` F1 declined on the selected subset, and numerous wrong predictions remained highly confident. These findings make abstention a reasonable subject for further citation-auditing research, not a finished safety mechanism. A stronger evaluation would use independent article-disjoint data, inspect

high-confidence failures against their evidence, and measure the consequences of missed citation errors in a realistic review workflow.

References

[1] Arman Cohan, Waleed Ammar, Madeleine van Zuylen, and Field Cady. 2019. Structural Scaffolds for Citation Intent Classification in Scientific Publications. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics*. 3586–3596. doi:10.18653/v1/N19-1361

[2] Yonatan Geifman and Ran El-Yaniv. 2017. Selective Classification for Deep Neural Networks. In *Advances in Neural Information Processing Systems*, Vol. 30. https://proceedings.neurips.cc/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html

[3] Steven A. Greenberg. 2009. How citation distortions create unfounded authority: analysis of a citation network. *BMJ* 339 (2009), b2680. doi:10.1136/bmj.b2680

[4] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On Calibration of Modern Neural Networks. In *Proceedings of the 34th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 70)*. 1321–1330. https://proceedings.mlr.press/v70/guo17a.html

[5] Maria Janina Sarol, Shufan Ming, Shruthan Radhakrishna, Jodi Schneider, and Halil Kilicoglu. 2024. Assessing citation integrity in biomedical publications: corpus annotation and NLP models. *Bioinformatics* 40, 7 (2024), btae420. doi:10.1093/bioinformatics/btae420

[6] David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hannaneh Hajishirzi. 2020. Fact or Fiction: Verifying Scientific Claims. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 7534–7550. doi:10.18653/v1/2020.emnlp-main.609

[7] David Wadden, Kyle Lo, Lucy Lu Wang, Arman Cohan, Iz Beltagy, and Hannaneh Hajishirzi. 2022. MultiVerS: Improving scientific claim verification with weak supervision and full-document context. In *Findings of the Association for Computational Linguistics: NAACL 2022*. 61–76. doi:10.18653/v1/2022.findings-naacl.6